Aryan Mathur

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Education

- **Indian Institute of Technology Palakkad** *2022- present*
  Bachelor of Technology in Electrical Engineering with Minor in Data Science ; GPA: 8.02 Kerala, India
- **Little Angels School Sonipat** *2007-2021*
  Central Board of Secondary Education Class XII: 97.4%; Class X: 98.2% Haryana, India

Work Experience

- **Mercor** *Jan 2026 – Present*
  Software Engineer San Francisco, USA (Remote)
  – Contributing to large-scale backend systems, working with extensive and complex production codebases.
  – Implementing new features and resolving bugs across services written in Go and Python.
- **Accenture** *May 2025 - July 2025*
  Advanced Application Engineering Intern Mumbai, India
  – Built cloud-native Java applications with scalable REST APIs, integrated NLP services into existing systems.
  – Designed and deployed end-to-end software pipelines on AWS using SageMaker, S3, and Lambda and applied DevOps for automation of data workflows.
- **EasyAlgo** *May 2024 - August 2024*
  AI/ML Intern New Delhi, India
  – Engineered predictive **time-series models** for stock market using Python, TensorFlow, and statistical techniques.
  – Engineered **NLP-driven sentiment analysis** pipelines from financial news to generate trading signals and optimized data workflows with **SQL, efficient data structures, and algorithmic techniques**

Research Publications

- **Fight Scene Detection and Violence Highlight Generation System**
  – Architected a custom CNN–BiLSTM model for fight scene detection, achieving 93.5% accuracy, built a temporal highlight generation pipeline for summarizing violent segments in long-form videos. arXiv.
- **Adapting Interleaved Encoders with PPO for Language-Guided Reinforcement Learning in BabyAI**
  – Implemented **Perception–Decision Interleaving Transformer (PDiT)** with PPO to improve multimodal grounding. Achieved **42% lower reward variance** and faster convergence (**160k** steps vs baseline 200k). arXiv:2510.23148
- **Explainable Detection of AI-Generated Images**
  – Proposed an explainable image authenticity detector combining a lightweight CNN (*Faster-Than-Lies*) and *Qwen2-VL-7B* for artifact localization. Achieved **96.5% accuracy** on the extended CiFAKE dataset with **175 ms inference latency** on 8-core CPUs, enabling real-time edge deployment. arXiv:2510.23775.

Projects

- **VartalApp** *GitHub*
  Full-Stack Collaborative Classroom Application June 2025 – December 2025
  – Designed and built a **cross-device full-stack application** enabling screen sharing, video sharing, chat, collaborative canvas, AI assistance, cloud persistence, and file sharing for classroom environments.
- **Splitzy Pay**
  Full Stack Financial Management Application Feb 2025 – Apr 2025
  – Built a full-stack expense splitting and financial analytics application as a DBMS course project.
  – Implemented UPI-style transfers, expense groups, category budgets, future expense planning, and unified financial ledger tracking.

Technical Skills

- **Languages:** Python, C++, Scala, Java, MATLAB, JavaScript, SQL
- **Backend and Frameworks:** PostgreSQL, NoSQL, Swing, ReactJS, Microservices, Node.js, Tailwind, MySQL
- **AI/ML:** Deep Learning, LLMs, NLP, RAGs, Computer Vision, Reinforcement Learning , Transformers, Big Data, Vision–Language Models, TensorFlow, PyTorch, Scikit-learn , Machine Learning, Gen AI
- **DevOps & Cloud:** AWS (Lambda, S3, SageMaker), Azure DevOps, Kubernetes, CI/CD Git, Linux, DBMS

Achievements

- **Meta OpenEnv Hackathon 2026**, Finalist
- **Adobe AI Challenge 2024**, Ranked 9th among all IITs
- **Flipkart GRiD 7.0 Hackathon** , National Semi-Finalist
- **Amazon ML Challenge 2025** , Top 0.9% , Ranked **251/27,600**+
- **National Physics Olympiad** , AIR 433 (Top 0.2%)

Positions of Responsibility

- **Placement Coordinator**, CDC, IIT Palakkad *May 2025- Present*
- **Founder and President**, Entrepreneurship Cell, IIT Palakkad *August 2024 - May 2025*
- **Contingent Leader**, Inter IIT Tech Meet 13.0 @ IIT Bombay *June 2024 - December 2024*